Final Check-Your-Understanding Questions

Based on Sections 1, 4.3, 7.2 and 7.4 of the uploaded measles modelling manuscript (25 October 2026).

Q1. Why does under-reporting change network size but not Rv?

Measure	South Carolina	Short Creek
Case-count adjustment	Doubled	Tripled
Network: original → adjusted	1,277 → 2,579	375 → 1,229
Rv: original → adjusted	2.40 → 2.42	1.64 → 1.63
h*: original → adjusted	58.3% → 58.6%	39.2% → 39.6%

Growth versus height. If a constant fraction rho of true cases is reported, observed incidence is Y(t) = rho I(t). For early exponential growth I(t) = I0 exp(rt), log Y(t) = log(rho I0) + rt. The multiplier changes the curve height, not its slope r or normalised shape. With the natural-history assumptions held fixed, the estimated reproduction number therefore changes little.

Network size. In an idealised closed outbreak, the final infected fraction z obeys z = 1 - exp(-Rv z) under standard homogeneous initial conditions. For fixed Rv, the fraction infected stays fixed, so larger total case counts require more initially susceptible people and a larger fitted effective network. The two-group leaky-vaccine model is more complicated, explaining why the network adjustment is only approximately proportional.

Referee concern. If reporting improves over time, Y(t) = rho(t) I(t), so d[log Y]/dt = d[log I]/dt + d[log rho]/dt. Improving ascertainment can create apparent extra growth, biasing the inferred growth rate, Rv, and derived thresholds. The manuscript's sensitivity result therefore addresses constant, not time-varying, under-reporting.

Q2. Three-sentence response to a referee

1. We thank the reviewer for this observation; however, González-Parra et al.'s R0 estimate of approximately 30–40 comes from a single-population early-growth model, whereas our Rv = 3.00 measures transmission in a partially immune, preferentially mixing network, so the two quantities are not directly comparable.

2. As discussed by Wallinga and Lipsitch, converting epidemic growth into a reproduction number depends on the assumed generation-interval distribution, which must be held fixed for a meaningful comparison.

3. Proposition 3 and our profile-likelihood analysis show that aggregate case counts cannot separately identify R0 and the mixing parameter theta, but can constrain the effective reproduction number and policy-relevant immunity thresholds through the identifiable combination R0 theta.

Supporting calculation. In the small-community limit, Rv approaches max(R0 s1, R0 theta s2). If the community dominates, Rv is approximately R0 theta s2. West Texas has R0 theta = 3.92 and Rv = 3.00, implying s2 approximately 3.00/3.92 = 0.765. The profile likelihood barely changes across R0 = 8–25, while theta compensates.

Q3. Two-sentence answer to the editor

Unlike previous single-population studies of these outbreaks, our study fits a two-group preferential-mixing model to three major US measles outbreaks and establishes analytically and through profile likelihoods that aggregate case data cannot separately identify the basic reproduction number and mixing intensity, while still allowing estimation of the effective reproduction number, network size, and policy-relevant immunity thresholds.

We also derive the shared critical transmission-blocking threshold $1 - 1/R_v$ for vaccination and isolation, and combine this result with dose-explicit optimal control and outbreak-specific counterfactuals to quantify how intervention strength, timing, and vaccine wastage affect outcomes.

Threshold check. Using rounded R_v values, West Texas: $1 - 1/3.00 = 66.7\%$; South Carolina: $1 - 1/2.40 = 58.3\%$; Short Creek: $1 - 1/1.64 = 39.0\%$. These are approximate limiting thresholds, not replacements for the manuscript's more precise fitted h^* values of 68.4%, 58.3%, and 39.2%.